Oscar I Hernandez

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SUMMARY

Scientific and financial researcher with experience in computer science, teaching, consulting, customer development, digital finance, and deep interest in differential topology and investment finance.

EXPERIENCE

Graduate Teaching Assistant
University of Alaska Fairbanks **August 2020 - Present, Fairbanks, AK**

- Lead two recitation sections for Calculus II (MATH 252L) for 50+ students.
- Assist students of all levels at the Math & Stat Tutoring Lab, 2 days a week.
- Grade weekly written homework for all synchronous sections of Calc II; previously a recitation TA/grader for Calculus in the Fall of 2020.

Founder & Electrical Engineer
ARKS **April 2018 - March 2020, Ontario, CA**

- Raised over $20k in grant funding from entrepreneurial programs including the National Science Foundation I-Corps, Blackstone Launchpad, YC Startup School, ExCITE Riverside, and the inaugural cohort of the Interphase Incubator.
- Built an Autonomous Mobile Robot prototype to improve warehouse picker productivity 2-3x by eliminating the long walk.

Full Stack Designer
Freelance Management Consulting **January 2018 - July 2020, Riverside, CA**

- Acquired 13+ clients and collaborated with 1-5 member teams to propose/design/develop/deploy full-stack technological solutions.
- Maintained 20+ websites for Columbia University, and many more for other clients.
- Supervised two summer interns, in collaboration with another consultant, on a $7k project.

PROJECTS

Research Publications
Bard College at Simon's Rock • August 2018

- Awarded Honors on my BA Senior Thesis for solving an open problem in theoretical computer science originally proposed in the 1970s.
- Received 20+ citations and 800+ reads on ResearchGate for academic articles published in the "European Journal of Combinatorics", the "Proceedings of the 4th International Conference on Applications in Nonlinear Dynamics" and the "12th Latin American Symposium, Ensenada, Mexico, April 11-15, 2016, Proceedings".
- Presented at the "SoCal NLP Symposium 2019" and the "AMS Fall 2019 Western Sectional Meeting" among other scientific venues.

EDUCATION

Master of Science in Mathematics
University of Alaska Fairbanks • Fairbanks, AK • May 2022

Bachelor of Arts in Computer Science and Mathematics
Bard College at Simon's Rock • Great Barrington, MA • 2017

INVOLVEMENT

Finance Committee Chair
University of Alaska Fairbanks • Associated Students of the UAF • September 2020 - Present

- Represent student body of 20,000+ as an elected senator, seat H, in the ASUAF senate, rules, and finance committees.
- Operate ~$140k+ annual budget via a combination of written legislation, funding request administration, financial planning & analysis.

SKILLS

Quantitative and Qualitative Analysis

Project Decomposition and Management

Problem Solving & Generalist Software Engineering

Proactive, Independent Research & Clear Communication